Eficacia de la azitromicina en EPOC

Una visión personal

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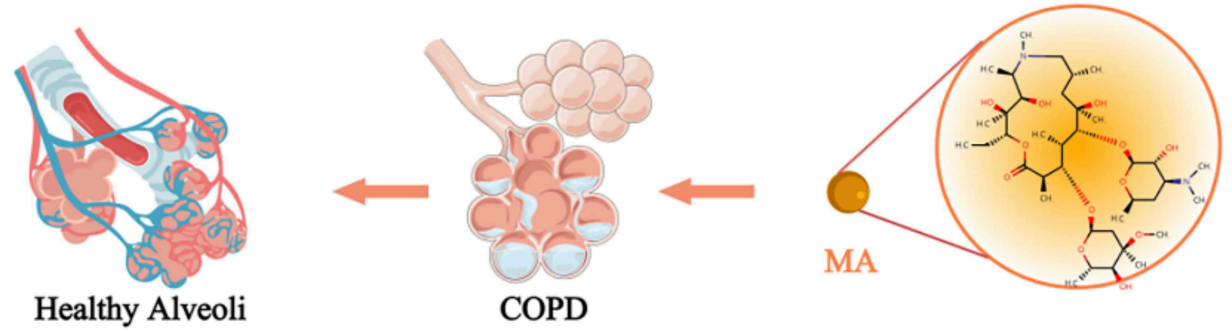
Conflictos de intereses

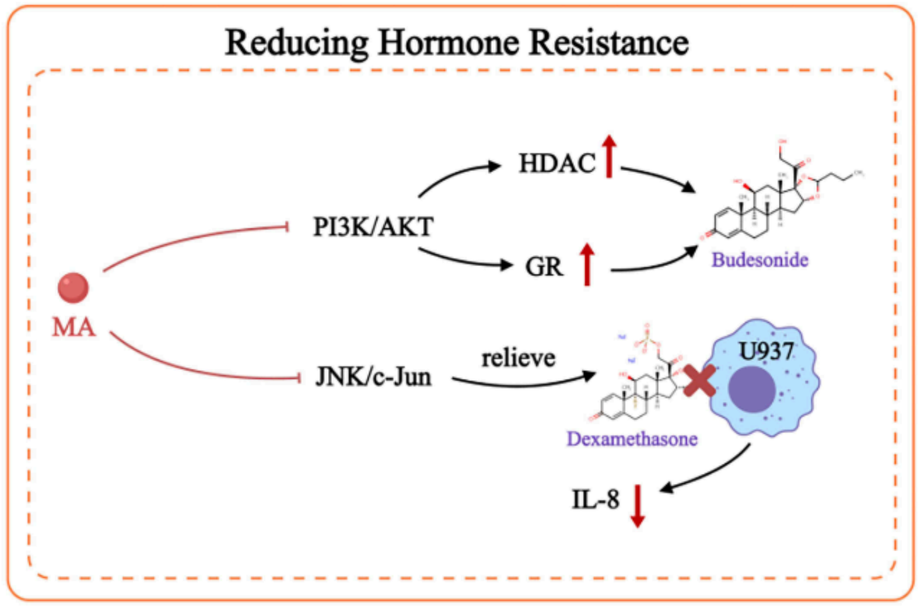
2021-2026

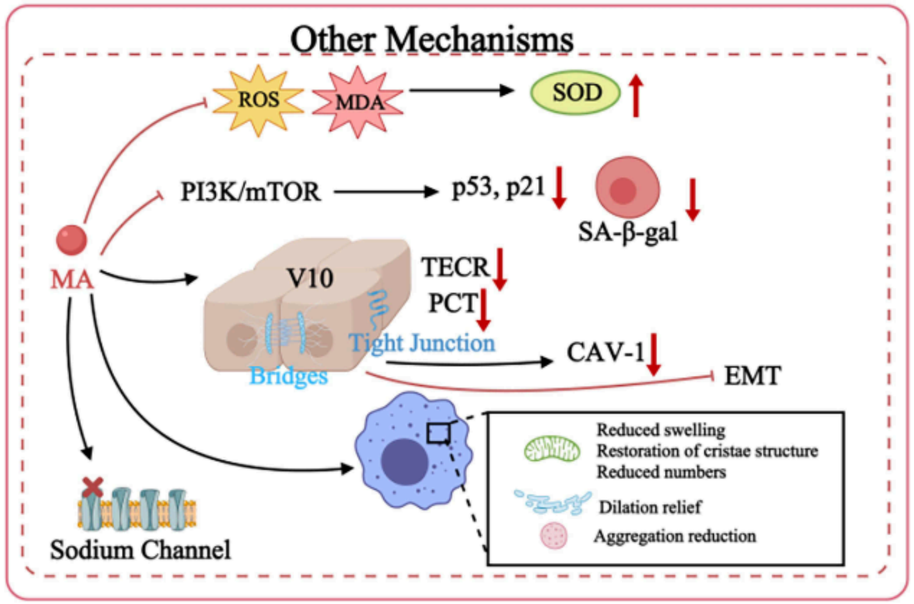
- **Pagos por conferencias/ asistencia a congresos/ asesoría científica:**
  - AstraZeneca/ Bial/ Chiesi/ CSL/ Grifols/ GSK/ MSD/ PulmonX/ Sanofi/ Zambon
- **Acciones o relaciones familiares con industria farmaceútica:** NO
- **Relación con industria del tabaco:** NO
- **Participación en guías de práctica clínica:** Comité Científico de GesEPOC
- **Patentes:** P201730724 (registrada)
- **Presidente electo de la SEPAR**

Azitromicina en el tratamiento de la EPOC

Posibles mecanismos



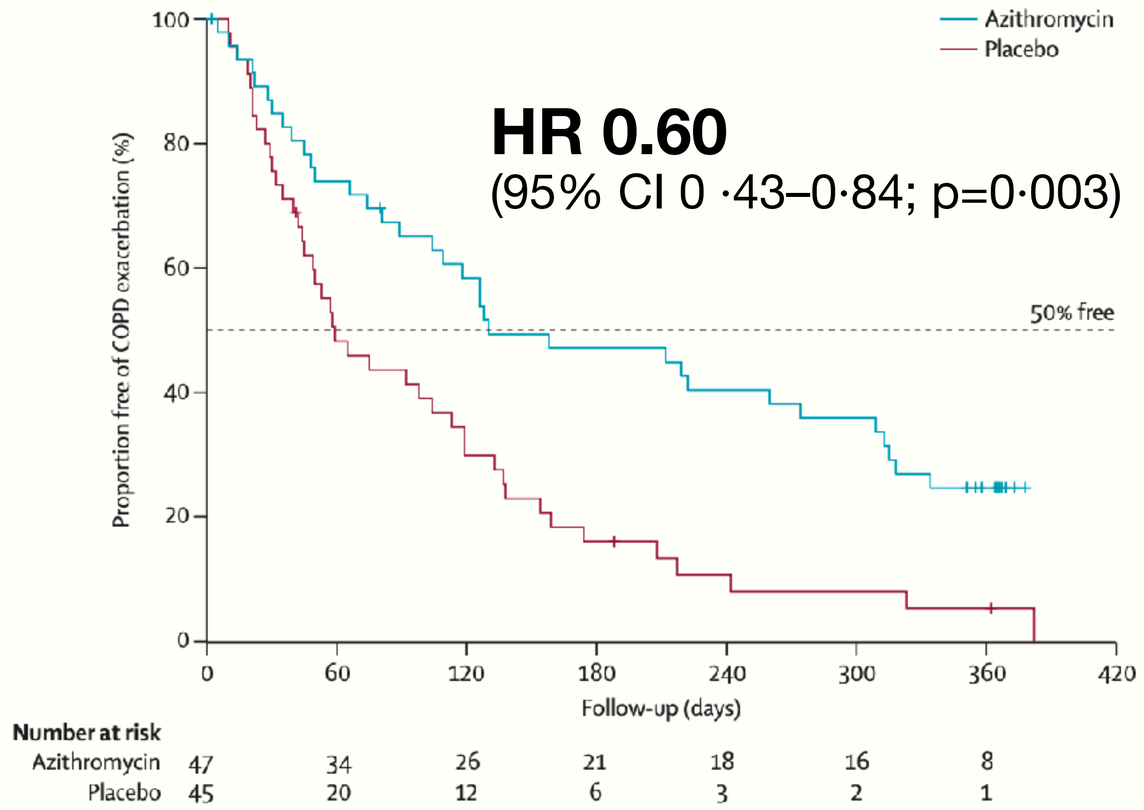




Wu J, Wu D, Wang H, Liu J, Chen Y. Mechanisms, Efficacy, Safety, and Pharmacoeconomics of Low-Dose, Long-Term Macrolide Antibiotics in the Treatment of Chronic Obstructive Pulmonary Disease: A Review. Int J Chron Obstruct Pulmon Dis. 2026;21:561395. Published 2026 Jan 29. doi:10.2147/COPD.S561395

Azitromicina en ensayos clínicos

Azitromicina 500 mg/48 h

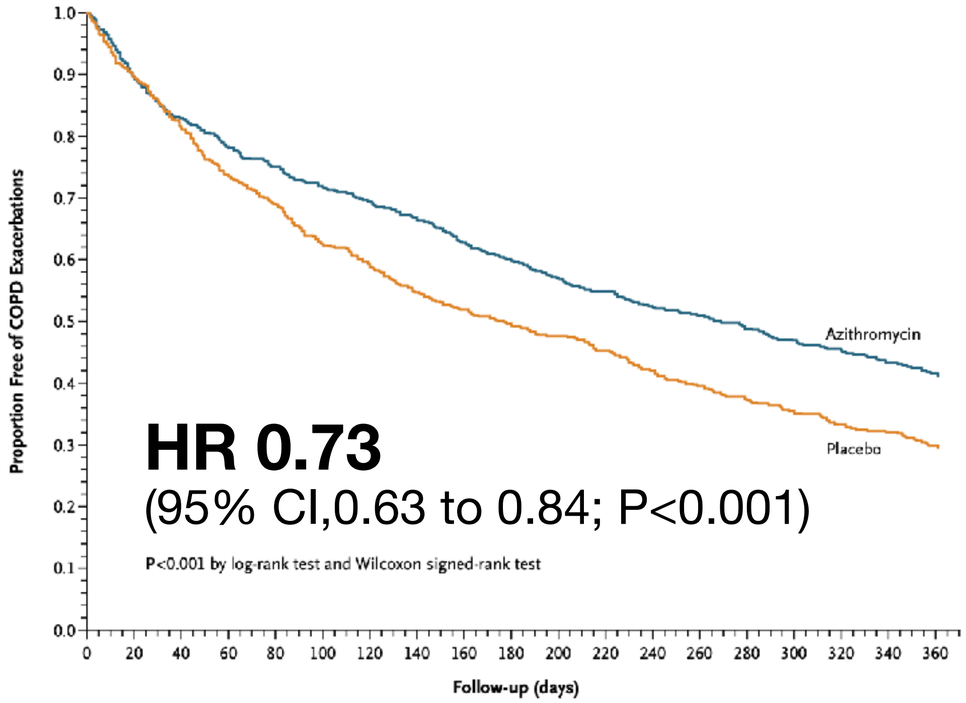


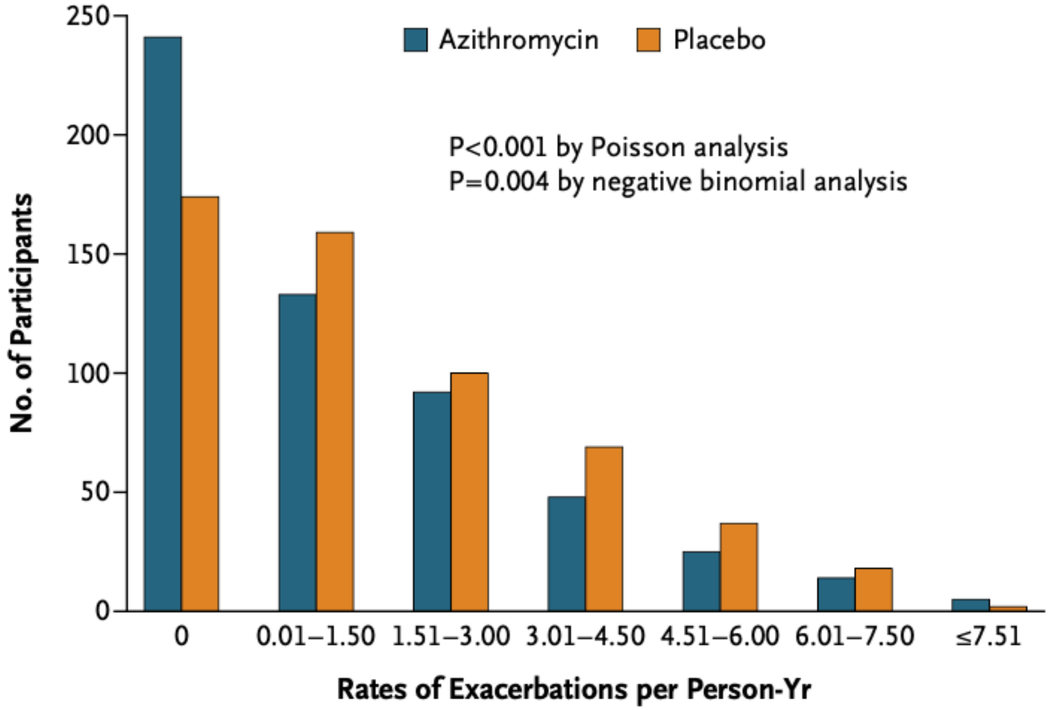
	At 12 months		Change from baseline at 12 months*			
	Azithromycin (n=41)	Placebo (n=36)	Azithromycin (n=47)	Placebo (n=45)	Difference (95% CI)	p value
Spirometry after bronchodilation						
FEV1 (L)	1·1 (0·47)	1·0 (0·42)	−0·03	−0·07	0·03 (−0·04 to 0·11)	0·37
FEV1 (% of predicted)	43·4 (17·9)	44·2 (20·1)	−1·13	−1·80	0·67 (−2·36 to 3·71)	0·66
FVC (L)	2·9 (0·93)	2·7 (0·79)	−0·04	−0·12	0·08 (−0·09 to 0·25)	0·35
FVC (% of predicted)	91·0 (23·5)	88·9 (20·9)	−0·73	−1·21	0·48 (−4·86 to 5·82)	0·86
6 min walk test (m)	415 (108)	379 (121)	−1·5	−20·8	19·3 (−17·8 to 56·5)	0·31
6 min walk test (% of predicted)	82 (20)	76 (23)	0·42	−3·55	3·97 (−3·66 to 11·60)	0·31
SGRQ total score	56·2 (17·2)	57·3 (15·2)	−1·05	−0·44	−0·61 (−5·75 to 4·53)	0·82
Symptoms	57·3 (18·0)	63·0 (14·4)	−4·97	1·80	−6·77 (−14·22 to 0·67)	0·075
Activity	75·5 (22·4)	76·1 (19·9)	−1·66	−1·37	−3·02 (−8·72 to 2·67)	0·30
Impacts	44·6 (17·8)	44·5 (18·3)	1·12	−1·19	2·31 (−4·43 to 9·05)	0·50
SF-12						
Physical component score	32·3 (10·7)	32·7 (10·3)	−0·76	1·13	−1·89 (−6·13 to 2·36)	0·38
Mental component score	36·8 (11·7)	35·9 (13·1)	−0·04	−1·80	1·76 (−4·02 to 7·53)	0·55
C-reactive protein (mg/L)	2 (1–30)	3 (1–90)	−20·6%	−2·1%	−18·9 (−50·6 to 33·2)	0·41
Leucocytes (×10⁹/L)	8·5 (3·1–16·2)	8·9 (4·8–16·3)	2·6%	9·9%	−6·7 (−17·2 to 5·1)	0·25

Uzun S, Djamin RS, Kluytmans JA, et al. Azithromycin maintenance treatment in patients with frequent exacerbations of chronic obstructive pulmonary disease (COLUMBUS): a randomised, double-blind, placebo-controlled trial. Lancet Respir Med. 2014;2(5):361-368. doi:10.1016/S2213-2600(14)70019-0

Azitromicina en ensayos clínicos

Azitromicina 250 mg/d





Albert RK, Connett J, Bailey WC, et al. Azithromycin for prevention of exacerbations of COPD. *N Engl J Med*. 2011;365(8):689-698. doi:10.1056/NEJMoa1104623

Azitromicina en ensayos clínicos

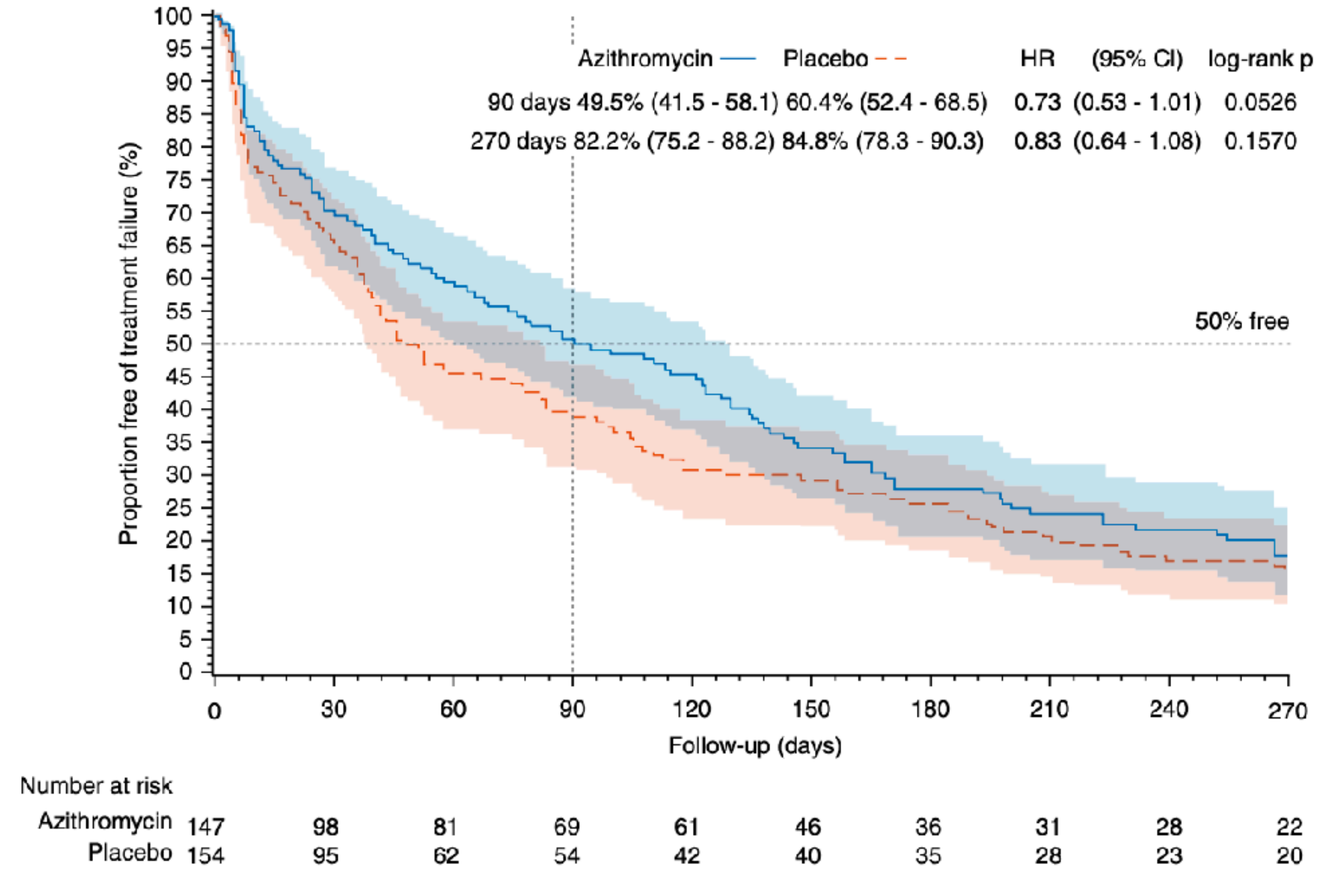
Azitromicina 250 mg/d

- **Seguridad en Albert et al, NEJM 2011**
  - Mortalidad por cualquier causa 3% azitromicina VS 4% en placebo (p=0.87).
  - Muerte por causa respiratoria 2% Vs 1% (p=0.48)
  - Muerte por causas CV 0.2% en ambos grupos
  - Pérdida de audición 25% Azitro vs 20% placebo (p=0.04)
  - Prevalencia de resistencias a macrólidos 52% en azitro y 57% en placebo (p=0.64)

Albert RK, Connett J, Bailey WC, et al. Azithromycin for prevention of exacerbations of COPD. *N Engl J Med*. 2011;365(8):689-698. doi:10.1056/NEJMoa1104623

Azitromicina en ensayos clínicos

Azitromicina 500 mg/d x3 días y posterior 250mg/48 h x 3 meses



Vermeersch K, Gabrovska M, Aumann J, et al. Azithromycin during Acute Chronic Obstructive Pulmonary Disease Exacerbations Requiring Hospitalization (BACE). A Multicenter, Randomized, Double-Blind, Placebo-controlled Trial. Am J Respir Crit Care Med. 2019;200(7):857-868. doi:10.1164/rccm.201901-0094OC

Azitromicina en RS y Metaanálisis

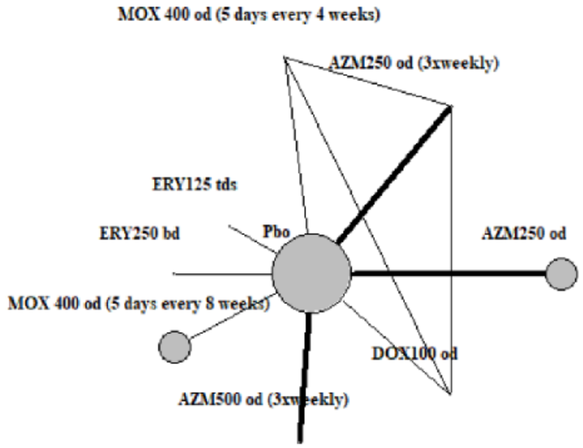
Revisión sistematica Cochrane 2021

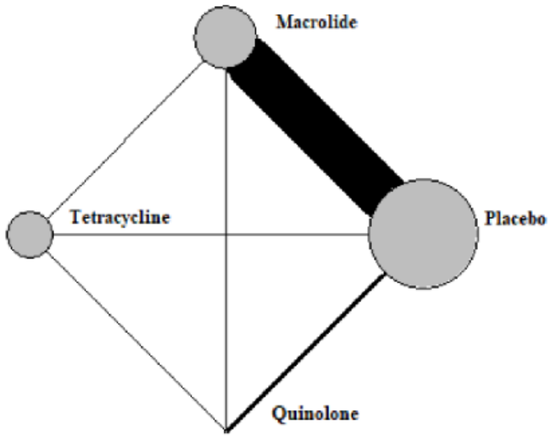
Treatment	Anticipated absolute effects (95% CrI)*		Relative effect HR (95% CrI)	No. of participants (studies)
	Absolute rate of exacerbations: median (95% CrI)	Risk difference with treatment (number of people experiencing exacerbations)		
Macrolide (weighted mean 50 weeks' duration)	1.34 (1.19 to 1.50)	127 fewer per 1000 (168 fewer to 87 fewer)	0.67 (0.60 to 0.75)	688 (6)
Tetracycline (13 weeks' duration)	2.58 (1.33 to 4.81)	60 more per 1000 (129 fewer to 127 more)	1.29 (0.66 to 2.41)	25 (1)
Quinolone (weighted mean 46.5 weeks' duration)	1.77 (1.50 to 2.08)	35 fewer per 1000 (87 fewer to 11 more)	0.89 (0.75 to 1.04)	594 (2)

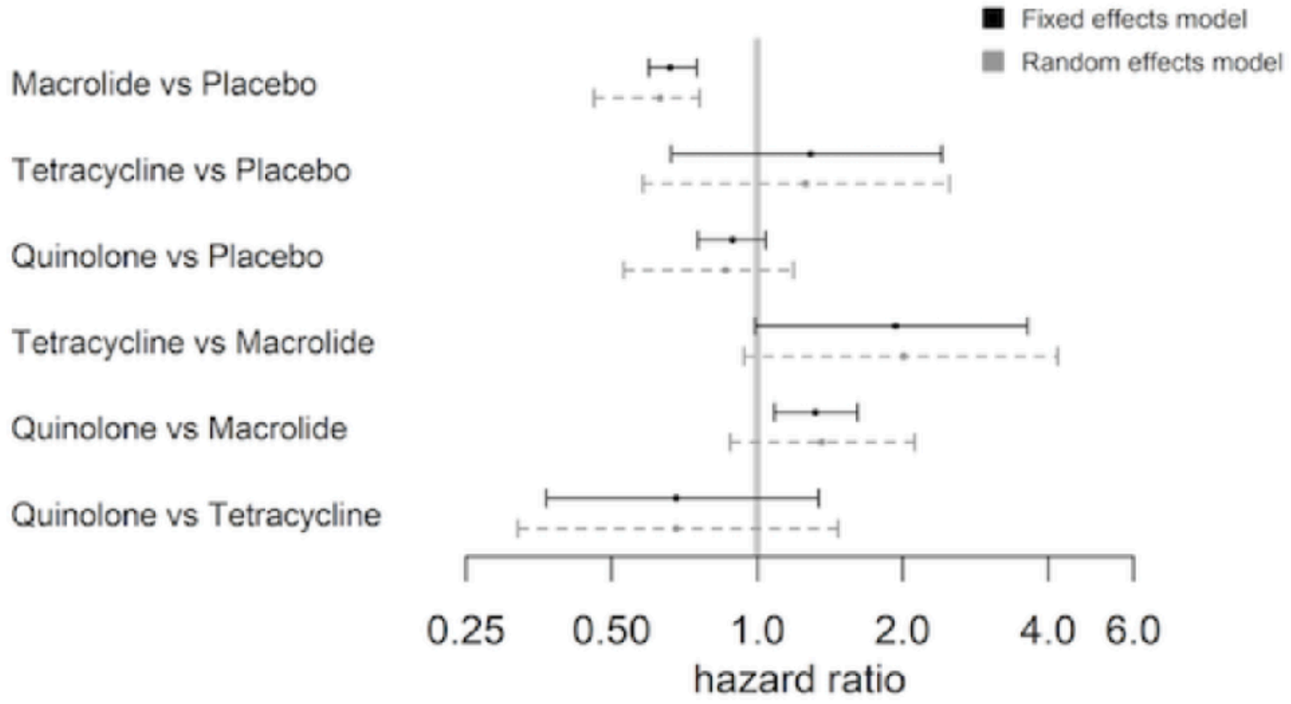
Janjua S, Mathioudakis AG, Fortescue R, et al. Prophylactic antibiotics for adults with chronic obstructive pulmonary disease: a network meta-analysis. Cochrane Database Syst Rev. 2021;1(1):CD013198. Published 2021 Jan 15. doi:10.1002/14651858.CD013198.pub2

Azitromicina en RS y Metaanálisis

Revisión sistematica Cochrane 2021







Janjua S, Mathioudakis AG, Fortescue R, et al. Prophylactic antibiotics for adults with chronic obstructive pulmonary disease: a network meta-analysis. Cochrane Database Syst Rev. 2021;1(1):CD013198. Published 2021 Jan 15. doi:10.1002/14651858.CD013198.pub2

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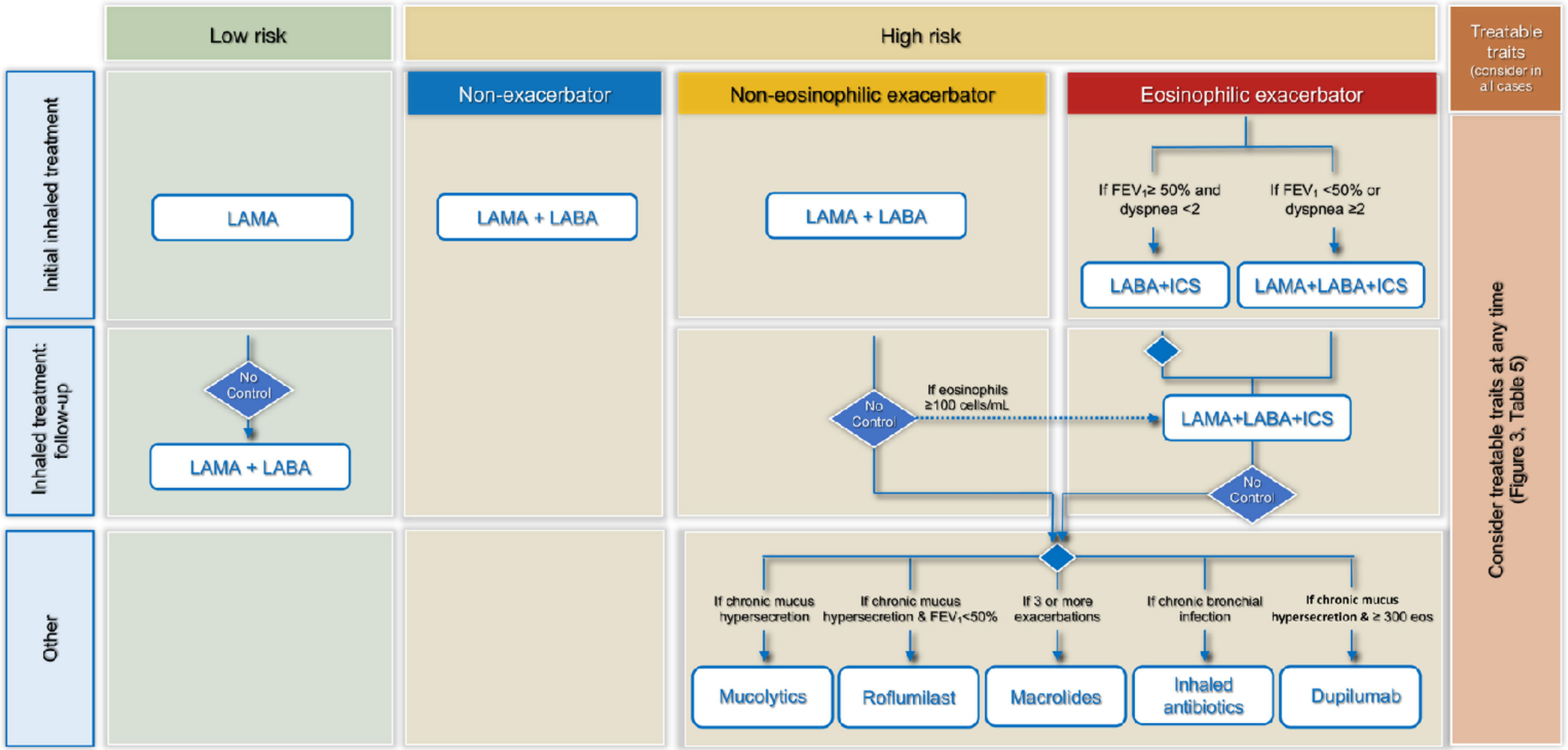
Revisión sistematica Cochrane 2021

Treatment	Anticipated absolute effects (95% CrI)*		Relative effect OR (95% CrI)	No. of participants (studies)
	Absolute probability of an SAE: median (95% CrI)	Risk difference with treatment*		
Macrolide (weighted mean 49 weeks' duration)	0.21 (0.18 to 0.25)	49.07 fewer per 1000 (81.18 fewer to 14.23 fewer)	0.76 (0.62 to 0.93)	971 (8)
Quinolone (48 weeks' duration)	0.26 (0.20 to 0.32)	1.873 fewer per 1000 (57.88 fewer to 60.89 more)	1.00 (0.72 to 1.34)	569 (1)
Macrolide + tetracycline (12 weeks' duration)	0.25 (0.15 to 0.37)	9.461 fewer per 1000 (1.07 fewer to 108.5 more)	0.97 (0.52 to 1.66)	101 (1)

Janjua S, Mathioudakis AG, Fortescue R, et al. Prophylactic antibiotics for adults with chronic obstructive pulmonary disease: a network meta-analysis. Cochrane Database Syst Rev. 2021;1(1):CD013198. Published 2021 Jan 15. doi:10.1002/14651858.CD013198.pub2

Azitromicina en GPCs y Recomendaciones

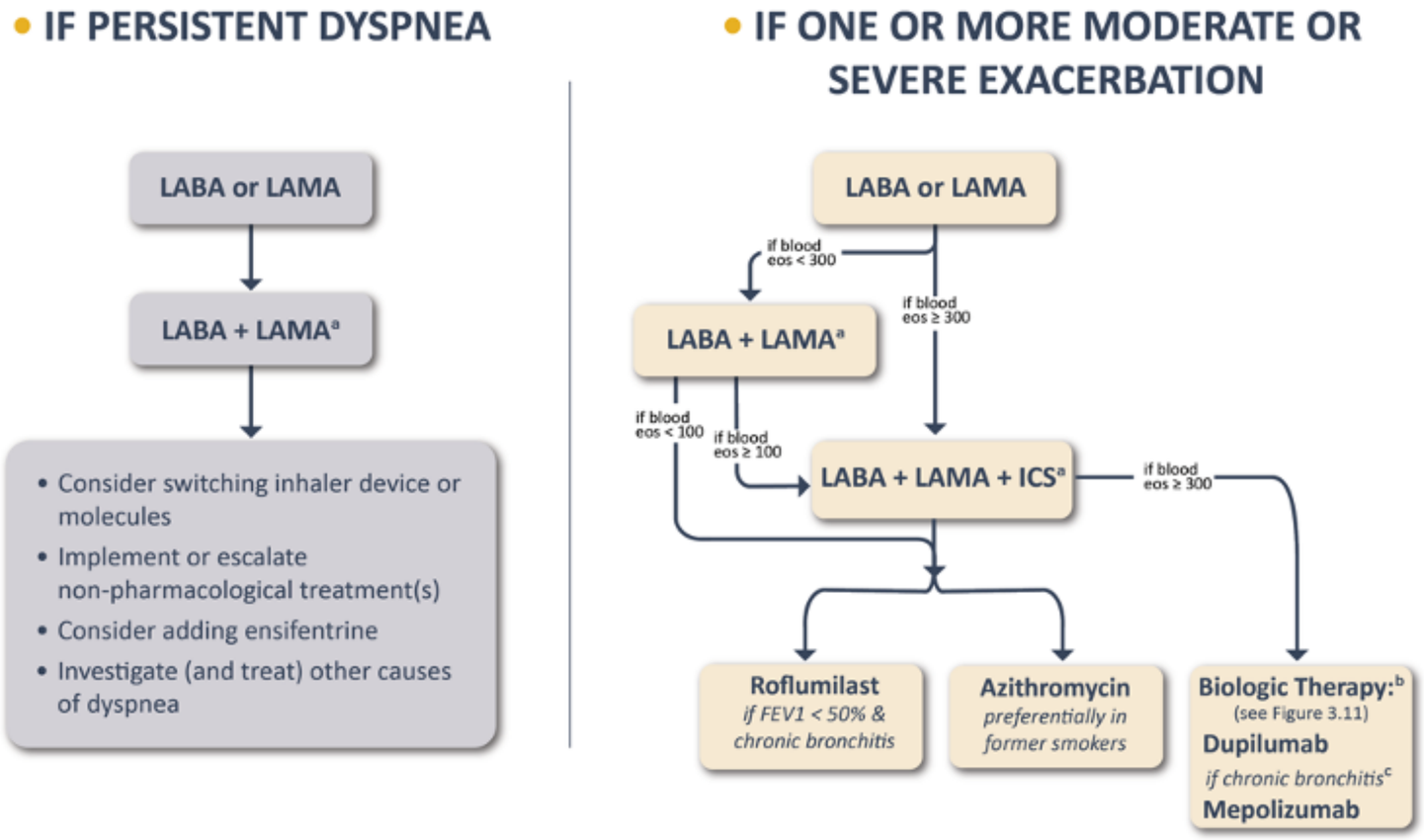
GesEPOC 2025



Miravitlles, M., Calle, M., Molina, J., Almagro, P., Gómez, J.-T., Trigueros, J. A., Sánchez-Angarita, E., Cosío, B. G., Casanova, C., Luis López-Campos, J., Riesco, J. A., Simonet, P., Rigau, D., Iskakova, A., Sanz, M. P., Sobradillo, P., Alcázar-Navarrete, B., Garin, N., & Soler-Cataluña, J. J. (2025). Update 2025 of the Spanish COPD Guidelines (GesEPOC). Pharmacological treatment of stable COPD◊. Archivos de Bronconeumología. https://doi.org/10.1016/J.ARBRES.2025.10.008

Azitromicina en GPCs y Recomendaciones

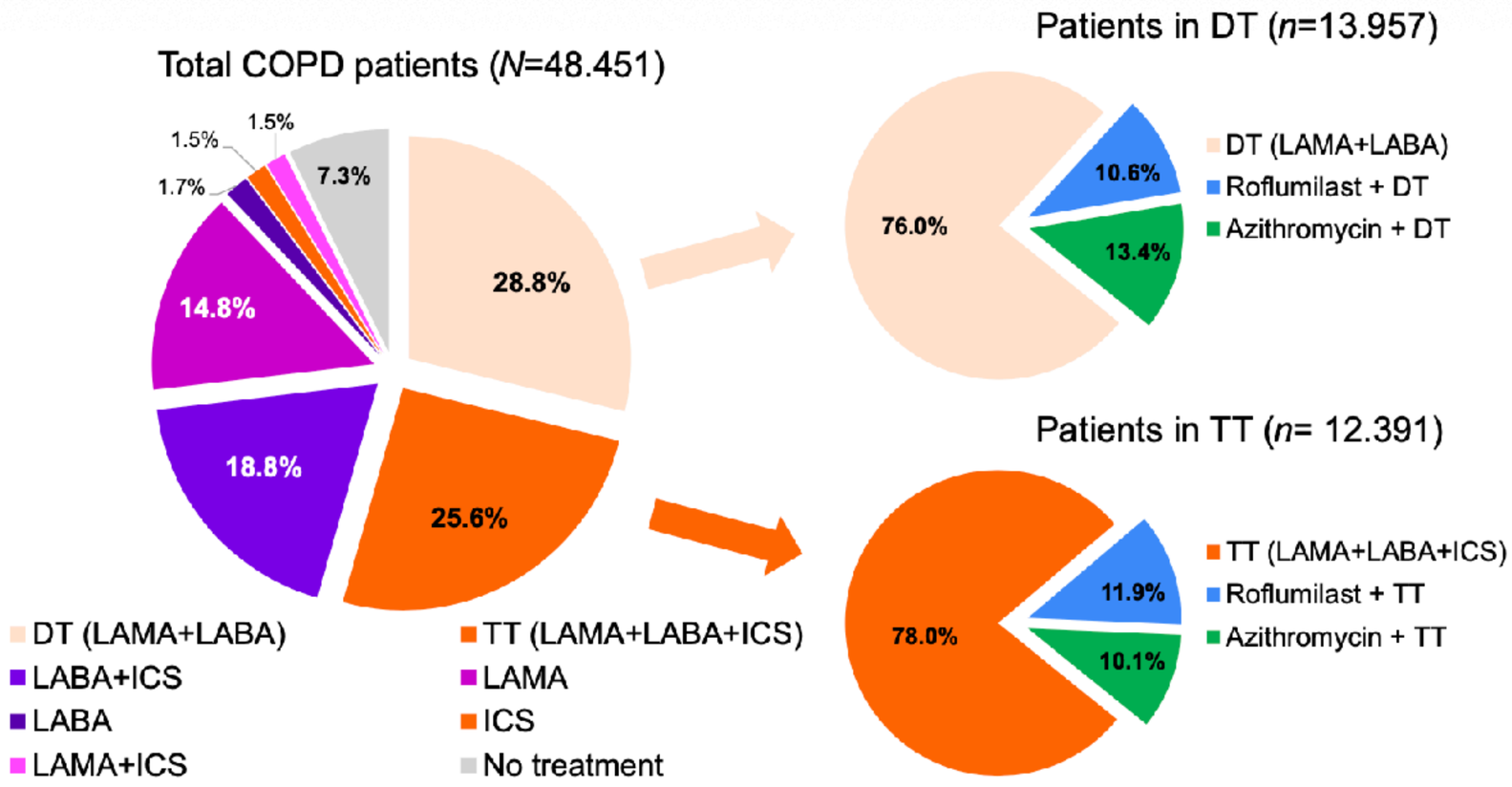
GOLD 2025



https://goldcopd.org/2026-gold-report-and-pocket-guide/

Azitromicina en la vida real

Resultados del estudio BREATHE-In en España



Navarrete BA, Rubio MC, Espín J, et al. A Real-World Perspective on Prevalence and Management of COPD in Spain: BREATHE-IN Study. ERJ Open Res 2026; in press (https://doi.org/10.1183/23120541.00400-2026).

Azitromicina en la vida real

Comparativa frente a iPDE4

Table 2 | Outcome effect estimates in new users of azithromycin versus roflumilast by matched cohort

	1:1 propensity score matched cohort			1:1 hdPS matched cohort		
	Events/person year			Events/person year		
Outcome	Roflumilast	Azithromycin	Hazard ratio (95% CI)	Roflumilast	Azithromycin	Hazard ratio (95% CI)
Moderate or severe COPD exacerbation	1.3	1.1	0.83 (0.79 to 0.87)	1.3	1.1	0.85 (0.81 to 0.90)
Moderate COPD exacerbation	1.1	0.9	0.83 (0.79 to 0.87)	1.1	0.9	0.82 (0.77 to 0.87)
Severe COPD exacerbation	0.2	0.2	0.93 (0.84 to 1.04)	0.2	0.2	1.01 (0.89 to 1.14)
All cause mortality	0.1	0.1	1.06 (0.93 to 1.21)	0.1	0.1	1.05 (0.90 to 1.23)
All cause hospital admission or death	0.7	0.6	0.94 (0.88 to 0.99)	0.7	0.6	0.93 (0.87 to 1.00)

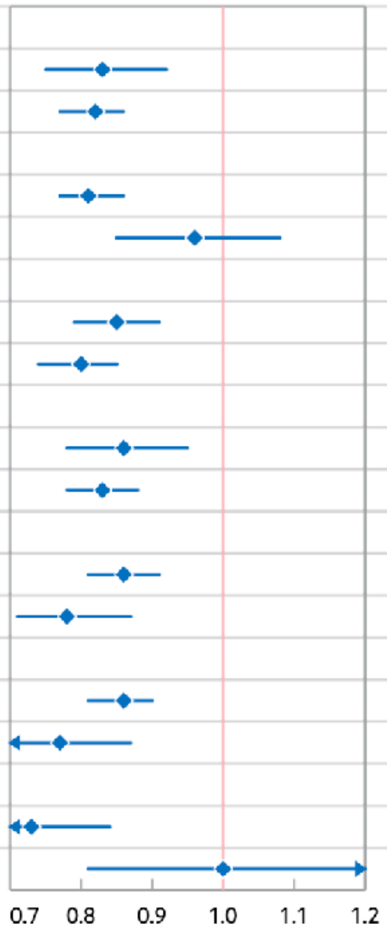
CI=confidence interval; COPD=chronic obstructive pulmonary disease; hdPS=high dimensional propensity score.

Portela GT, Wang SV, Suissa S, Schneeweiss S, Feldman WB. Comparative effectiveness of azithromycin versus roflumilast in patients with chronic obstructive pulmonary disease: emulated target trial. BMJ. 2026;394:e100111. Published 2026 Aug 5. doi:10.1136/bmj-2026-100111

Azitromicina en la vida real

Comparativa frente a iPDE4

Subgroup	Hazard ratio (95% CI)	Interaction P value	Interaction Hazard ratio (95% CI)
Baseline COPD exacerbation		0.82	
0	0.83 (0.75 to 0.92)		Reference
>= 1	0.82 (0.77 to 0.86)		0.99 (0.88 to 1.11)
Baseline COPD hospital admission		0.04	
0	0.81 (0.77 to 0.86)		Reference
>= 1	0.96 (0.85 to 1.08)		1.14 (1.01 to 1.30)
Any past asthma diagnosis		0.18	
No	0.85 (0.79 to 0.91)		Reference
Yes	0.80 (0.74 to 0.85)		0.94 (0.85 to 1.03)
Inhaled triple therapy		0.91	
No	0.86 (0.78 to 0.95)		Reference
Yes	0.83 (0.78 to 0.88)		0.99 (0.89 to 1.11)
Continuous inhaled triple therapy (90 days)		0.89	
No	0.86 (0.81 to 0.91)		Reference
Yes	0.78 (0.71 to 0.87)		1.01 (0.90 to 1.13)
Index prescription from pulmonologist		0.73	
No	0.86 (0.81 to 0.90)		Reference
Yes	0.77 (0.69 to 0.87)		0.98 (0.86 to 1.11)
Baseline eosinophils (cells/µL)		0.17	
≤300	0.73 (0.64 to 0.84)		Reference
>300	1.00 (0.81 to 1.24)		1.20 (0.92 to 1.57)



Conclusiones

- El empleo de azitromicina añadido al tratamiento de mantenimiento del paciente con EPOC se asocia a mejoría en el numero de exacerbaciones, sin cambios en la función pulmonar.
- Los estudios han incluido dosis diferentes, pero las más utilizadas son 250 mg/d o 500 mg/ 48 h
- El perfil de seguridad de la azitromicina debe ser evaluado de forma individual por el riesgo de ototoxicidad y de alargamiento del QT.
- Azitromicina es probablemente superior a otras terapias orales para la EPOC como iPDE4 en la prevención de exacerbaciones.
- Antes de valorar el tratamiento con macrolidos a largo plazo, se necesita una discusión con el paciente acerca del riesgo/ beneficio esperable.


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